

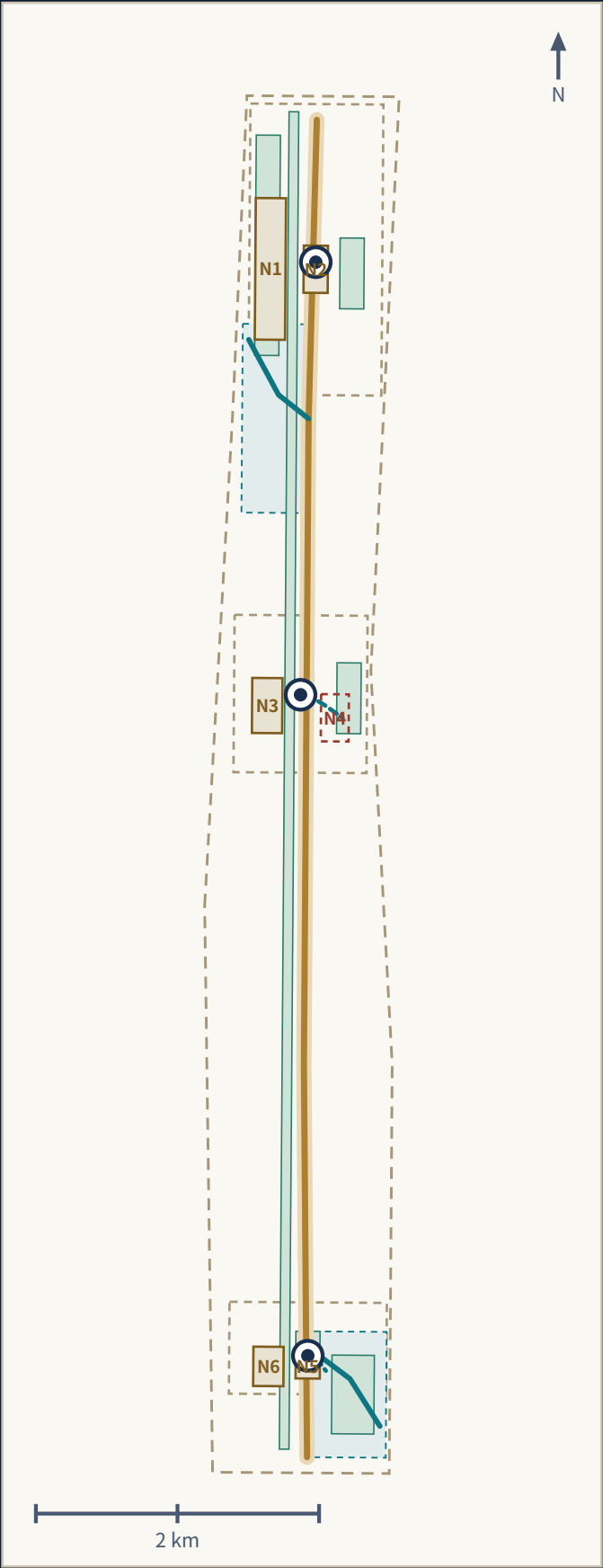
# HAIDIAN AI AXIS

## FROM COAL TO COMPUTE

Centennial Jing-Zhang AI Innovation Belt international urban design open call • open-source agent submission (concept proposal; not a statutory plan and not a government-approved name)

Heritage spine → three-station roles → two-wing supply/feedback → six spatial prototypes → ten operating scenarios → one annual ledger

|               |                                                                |
|---------------|----------------------------------------------------------------|
| Document      | Page • 15 pages                                                |
| Submission    | submissions/harry-water/haidian-ai-axis                        |
| Package state | scaffold (SCAFFOLD-DRAFT retained)                             |
| Metrics       | 14 known / 23 unknown                                          |
| Boundary      | official_boundary=false • boundary_precision=provisional_rough |



A single reading spine: every chapter is one link on it

- 1

**Heritage spine**  
First a continuous, free, exitable public space
- 2

**Three-station roles**  
Exchange / verify / launch — differentiated, not three identical demo zones
- 3

**Two-wing supply and feedback**  
Factor supply and real-demand stress testing
- 4

**Six spatial prototypes**  
A combinable grammar of gateway, courtyard, station, market, garden, stitch
- 5

**Ten operating scenarios**  
Every card has human review, an offline alternative and exit conditions
- 6

**One annual ledger**  
Checks back whether the first five links still hold

Non-negotiable factual and operational boundaries

- 1

HAIDIAN AI AXIS is not a statutory planning axis, not an extension of or second Beijing Central Axis, not a World Heritage–related project, and not a government-approved name.
- 2

The 1909 Jing-Zhang Railway connected Liucun in Fengtai, Beijing with Zhangjiakou and did not directly enter today's Inner Mongolia; that linkage belongs to the later Ping-Sui / Jing-Bao evolution chain.
- 3

The Haidian–Ulanqab twin-city compute coordination is this proposal's auditable dispatch model (proposal inference), not an approved or operating official architecture; no capacity, price, SLA or investment is promised.
- 4

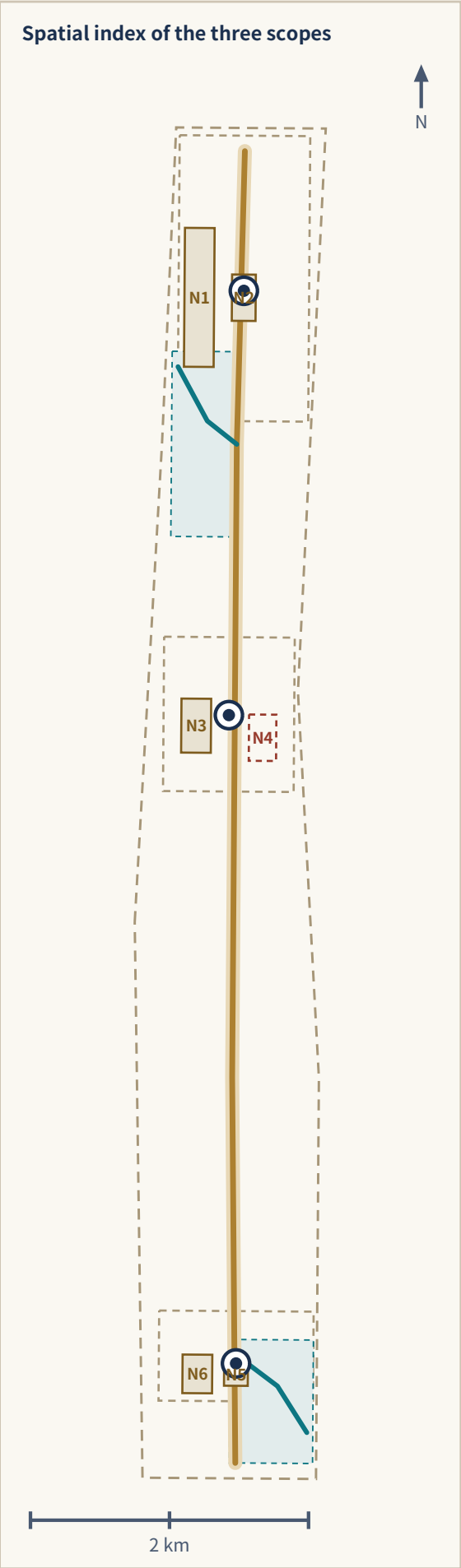
Every staffed desk, staffed attendance or human-callable point is a design target realizable during authorized service windows after an operator is confirmed; it is not existing staffing.

Four evidence tiers

| Tier                     | Content                                                                        | May support                                                       | Never                                                                 |
|--------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------------|-----------------------------------------------------------------------|
| E1 Official task basis   | Open-call announcement, agent taskbook                                         | Three scopes, three key areas, design tasks and output depth      | Never a precise redline or approval conclusion                        |
| E2 Public standards      | Urban design measures, regulatory plan measures, land-use classification guide | Urban-design depth, public-space requirements, land-use semantics | Never an approved control indicator                                   |
| E3 Public primary data   | Government statistics, bulletins, policy documents, official reporting         | Regional climate and compute-policy background, historical facts  | Never city-wide values as site values, or reporting as a measured SLA |
| E4 Project-side material | Middle-East case project sites and host-institution releases                   | Mechanism reference, operating organization                       | Never independent proof that performance has been delivered           |

Submitted evidence layer (this document adds no data)

| File                            | Features / content                                                                                       |
|---------------------------------|----------------------------------------------------------------------------------------------------------|
| geometry/site_boundary.geojson  | SITE-001                                                                                                 |
| geometry/key_areas.geojson      | PROV-KEY-001 • PROV-KEY-002 • PROV-KEY-003                                                               |
| geometry/land_use.geojson       | LU-001 • LU-002 • LU-003 • LU-004 • LU-005 • LU-006                                                      |
| geometry/buildings.geojson      | BLDG-ST01-001 • BLDG-ST02-001 • BLDG-ST03-001                                                            |
| geometry/roads.geojson          | ROAD-AXIS-001 • ROAD-WING-TECH-001 • ROAD-WING-XIAOYUE-001 • ROAD-STITCH-N4 • ROAD-STITCH-N5             |
| geometry/green_space.geojson    | GREEN-AXIS-001 • GREEN-QINGHE-001 • GREEN-XIAOYUE-001 • GREEN-ST01-001 • GREEN-ST02-001 • GREEN-ST03-001 |
| geometry/public_space.geojson   | N1 • N2 • N3 • N4 • N5 • N6                                                                              |
| geometry/constraints.geojson    | CONSTRAINTS • WING-TECH-001 • WING-XIAOYUE-001                                                           |
| geometry/phasing.geojson        | PHASE-001 • PHASE-002 • PHASE-003                                                                        |
| metrics.json                    | 14 known / 23 unknown                                                                                    |
| sources.json / assumptions.json | 42 sources • 18 assumptions                                                                              |



Three scopes: announced extent, work objective and output

| Scope / size                                  | Announced extent                                                                             | Work objective                                                                             | Position on the spine                                                          | Output                                                                                  |
|-----------------------------------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Coordinated research area<br>approx. 43.6 km² | North: North 5th Ring Rd; East: Jingzang Expwy;<br>South: Xizhimenwai St; West: Wanquanhe Rd | Energy–compute–knowledge network north of<br>Beijing, industry ecosystem and mechanism set | Regional background of the spine;<br>origin of two-wing supply and<br>feedback | Strategy diagram, mechanism chain,<br>scenario and operating framework                  |
| Overall design area<br>approx. 11.4 km²       | Urban and industrial areas 1–2 km around the<br>Jing-Zhang Heritage Park                     | Heritage spine, six spatial prototypes, renewal<br>framework and character control         | The heritage spine and the six spatial<br>prototypes                           | Land use, roads, green, public space,<br>buildings and phasing layers                   |
| Key detailed-design areas<br>approx. 368.4 ha | North to south: Zhongzhiyuan 192.1 ha, AI Origin<br>Community 104.3 ha, Dazhongsi 72.0 ha    | Detailed design of three differentiated urban<br>stations and their concept nodes          | Where the three-station roles and ten<br>scenarios land                        | Positioning, structure, renewal, active travel,<br>public space, AI scenarios and risks |

The three levels resolve one another; they are not three unrelated drawing sets

The coordinated research area answers what the northbound corridor exchanges in the AI era; the overall design area answers how that exchange becomes walkable, dwellable, exitable urban space; the key areas answer how the three stations carry exchange, verification and launch.

Rectangle edges are not parcel lines or rights-of-way, and the areas are a discussion basis only. Once official polygons are published, the bases of site\_area\_sqm, key\_area\_count, green\_ratio, public\_space\_ratio, building\_footprint\_area\_sqm and floor\_area\_ratio all change, and geometry generation, metric recomputation, figure rendering and self-check must all be rerun.

Provisional status: SITE-001 and PROV-KEY-001/002/003 are maintainer-registered provisional rough surfaces (official\_boundary=false, boundary\_precision=provisional\_rough). The thin dashed outline is a discussion extent only; it is not a parcel line, road right-of-way or any statutory control line. Every layer and metric must be recomputed together once official polygons are published.

From coal to compute: one northbound corridor, three eras of functional transformation

| Era              | What the northbound corridor carries           | What it gives Beijing / Haidian                    | What Haidian exports                                        |
|------------------|------------------------------------------------|----------------------------------------------------|-------------------------------------------------------------|
| Railway era      | Coal, goods, people                            | Industrial fuel, goods exchange, regional linkage  | Manufactured goods, markets and urban services              |
| Zhongguancun era | Talent, knowledge, equipment, capital          | Research and industrial innovation capacity        | Technology outputs, firms and services                      |
| AI era           | Compute tasks, models, data products, services | Suitable nearby green compute and elastic capacity | Algorithms, models, standards, talent, scenarios and demand |

This is a two-way exchange, not one-way delivery: while northern resources come south, Haidian's algorithms, models, standards, talent and scenario demand go north, and the axis turns both into public urban services. “From coal to compute” names the functional transformation of one and the same northbound regional exchange corridor across eras; it claims no shared physical alignment with the Jing-Zhang Railway. The 1909 line ended at Zhangjiakou, and the link toward Inner Mongolia belongs to the later Ping-Sui / Jing-Bao evolution chain.

Beijing Central Axis: three distinctive correspondences

| What is distinctive                                              | Correspondence                                     | Forbidden reading                                                |
|------------------------------------------------------------------|----------------------------------------------------|------------------------------------------------------------------|
| A composite of urban buildings and sites, not a road             | The spine is a set of spaces, not a line           | Never call it a northern extension or second central axis        |
| Nodes differ in function and order, balanced in pairs            | A differentiated node sequence with paired balance | Never copy palaces, gate towers, altars or their value hierarchy |
| Construction and everyday life of different eras keep overlaying | Continuous overlay of time and culture             | No static pseudo-historical theme park                           |

Three local planning inferences (contemporary practice, not unique to the Central Axis)

| Inference               | Spatial and operating action                                                                                                                                                                                                  | Forbidden reading                                                               |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| 1. Cross-axis stitching | Stitching across the railway, Qinghe and Xiaoyue blue-green cross-links, community transfer and east-west public space; spine and cross-links matter equally                                                                  | Drawing a line is not a finished design; feasibility must be studied separately |
| 2. Public sharing       | The spine and the three landmarks stay free, continuous and enterable, with non-digital entrances. Human service is a design target only in authorized windows after operator confirmation; no venue opens only on event days | Public openness is not an approved ownership or management arrangement          |
| 3. Long-term governance | From one-off construction to annual operation and review: the Axis Timetable, an open test season, the compute ledger, human review, public feedback and annual updates                                                       | Events and operating bodies are not approved government arrangements            |

Applicability limits of the Haidian–Ulanqab operating model

Fibre reaching Beijing / Yizhuang is not physical arrival in Haidian; the reported ~4.2 ms is a link figure in reporting, not an application-level SLA, and this proposal never writes “4.2 ms direct to Haidian” .

The PUE figures 1.32 / 1.22 belong to one specific data centre in the period described by the source; they do not represent the start-up zone, all of Ulanqab, or any current average.

A signed memorandum proves no capacity, price, SLA, contract or budget. As of 2026-08-08 no project-specific agreement or operating evidence has been provided to or independently verified by this submission.

Task tiering decides where computation happens: privacy-sensitive and on-site control never leave; low-latency public services run local or near-edge; ordinary inference and elastic peaks are dispatched under agreement; training and batch work prefer nearby partners; any failure falls back to local degradation and human takeover.

Privacy boundary

Data on identity, biometrics, minors, health and trajectory never enter outbound tasks; public services never require authorizing personal data; opting out never lowers basic public service.

Energy, carbon and water ledger

Per task, record latency, queueing, failure rate, energy, the applicable PUE definition, green-power share, carbon, water and switchover time, each with its metering definition and source. Until real operating data exist, only the test specification and an empty table structure are published.

Failover and annual review

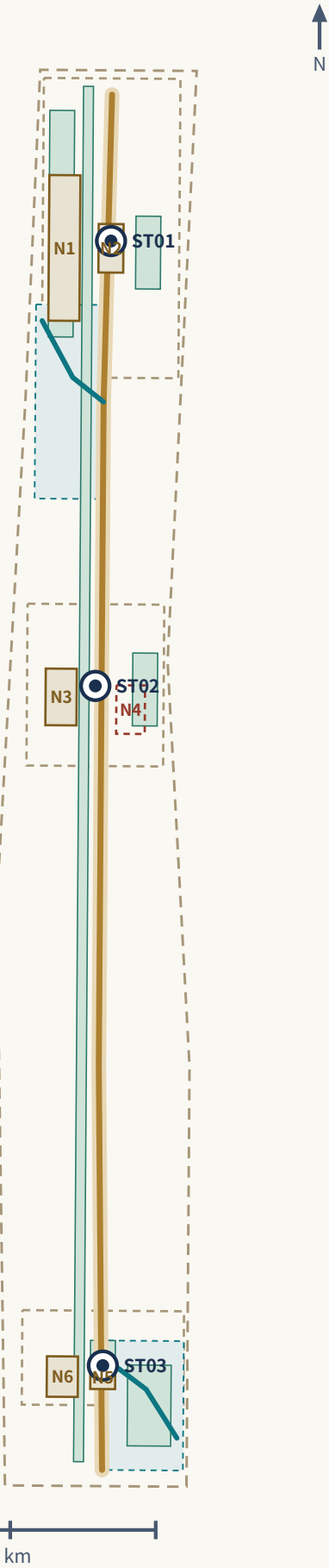
Any switchover can be vetoed by a person; human takeover is the default fallback, not an exception; every switchover and takeover enters the public ledger. One open test season and ledger review per year is published and can revise the task-tiering rules in return.

# One axis, three stations, two wings

The spine is first a free, continuous, accessible, climate-adapted public space and only then a display interface

HAIDIAN AI AXIS: FROM COAL TO COMPUTE  
Railway Heritage. AI Innovation. Everyday Life.

## One spine, three stations, two wings



### ST01 • Exchange and dispatch Zhongzhiyuan AI Independent Innovation Acceleration Area Compute Exchange

**Positioning**  
The carrier of full-stack independent innovation and the voice in AI governance; the only node on the new axis that carries the decision and display of “where does the task get computed.”

**Spatial structure**  
The Qinghe-front interface becomes the public living room, forming a layered “river—garden—courtyard—R&D cluster” structure; dispatch and testing functions concentrate in a visitable central courtyard group, avoiding a closed command center.

**Public space**  
The low-carbon innovation social environment centers on the Garden prototype, carrying shade, stormwater detention, and quiet rest; during public testing seasons some courtyards convert to open test grounds.

**AI scenarios**  
Scenario cards 02 Safety Governance Sandbox, 06 Jing-Zhang Heritage and AI Public Education, 08 Haidian-Ulanqab Green Compute Dispatch Test, 10 Climate Digital-Twin City Operations Test.

**Concept nodes**  
N1 Qinghe Compute Commons — Waterfront linear park + a group of open courtyards  
N2 Fail-safe Test Yard — Hard-surfaced yard that can be closed or opened + outer gallery

### ST02 • Verification and incubation Beijing AI Origin Community Origin Commons

**Positioning**  
The validation and incubation area of the world-class AI innovation ecosystem; models and public services undergo dual review by the community and universities here before entering the city.

**Spatial structure**  
A small-scale network centered on the Courtyard prototype, stitching campus—park—neighborhood; open-source honor display is arranged linearly along the walking-and-cycling spine rather than concentrated in a single exhibition hall.

**Public space**  
The Origin Commons must be genuinely enterable and lingerable with human-staffed service — no giant-screen hall; it hosts an information and service interface for newcomers' initial arrival period (service hours and staffing are design targets to be verified, subject to operating conditions).

**AI scenarios**  
Scenario cards 01 Open-Source Launch Hall, 04 Newcomer 72-Hour Settling-In, 06 Jing-Zhang Heritage and AI Public Education, 09 Research-to-SME Product Conversion Test.

**Concept nodes**  
N3 Origin Courtyard — Semi-enclosed community courtyard + open street-level ground floors  
N4 Campus-Station Stitch — A walking-and-cycling stitch across a barrier; form intent pending study

### ST03 • Launch and translation Dazhongsi AI Industry Cluster Axis Launch Station

**Positioning**  
The launch and conversion area for intelligence-native new business forms; the new axis's outward interface to the market, the public, and the world.

**Spatial structure**  
Led by the Market prototype, organizing four-quadrant pedestrian connectivity and ground-floor commercial services around the rail station; launch and cultural-content functions string along the pedestrian system.

**Public space**  
The Axis Launch Station is public space, not a corporate showroom; it is the main venue of the annual “New Axis Timetable” while remaining an everyday urban living room usable by ordinary citizens.

**AI scenarios**  
Scenario cards 05 Continuous Barrier-Free Passage, 07 Night-Shift Safe Journey Home, 03 Network/Power Outage Fail-Safe, 10 Climate Digital-Twin City Operations Test (launch end).

**Concept nodes**  
N5 Four-Quadrant Crossing — Continuous walking surfaces and waiting spaces at four corners  
N6 New Axis Launch Market — Adaptable open street frontage + a permanent canopy

## Six urban spaces: a repeatable, combinable block grammar

| Space type | Urban role                                              | Typical design action                                                                                                                                   | Layer          |
|------------|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Gateway    | Information, accessibility and public-service interface | Station entrances, wayfinding and accessible ramps; a staffed desk is a design target for authorized windows; no pseudo-historical gate tower           | N5             |
| Courtyard  | Exchange, community courses, rest and small trials      | Semi-open courtyard, bookable test bays, shade and seating                                                                                              | LU-004         |
| Station    | Exchange node for compute, mobility, service or events  | Edge compute point, transfer and event gathering; a non-digital entrance is always kept, while human service is only an authorized-window design target | ROAD-AXIS-001  |
| Market     | Showcase, procurement, culture and everyday commerce    | Open street frontage, activated ground floor, temporary exhibits                                                                                        | LU-003         |
| Garden     | Shade, winter sun, stormwater detention and quiet rest  | Climate public space, floodable green, rain gardens                                                                                                     | GREEN-AXIS-001 |
| Stitch     | East-west links across rail, road and water barriers    | Active-travel stitching and ecological links; feasibility studied separately                                                                            | ROAD-STITCH-N4 |

The six spaces are not a fixed building list but combinable prototypes offered to the professional teams that follow; they are concept proposals. The spine is first a free, continuous, accessible, climate-adapted public space and only then a display interface, and no smart facility may be placed at the cost of continuity, shade, rest or accessibility.

## Two wings and the coordination loop

**Zhongguancun technology-service wing**  
Factor supply: capital, legal, IP, talent and international networks

**Xiaoyue River scenario-enablement wing**  
Real-demand feedback: stress-testing community, ecology, mobility and everyday needs

Loop direction: Origin Community verifies → Zhongzhiyuan exchanges and dispatches → Dazhongsi launches and translates → the two wings supply factors and real demand → back to Origin Community for the next round.

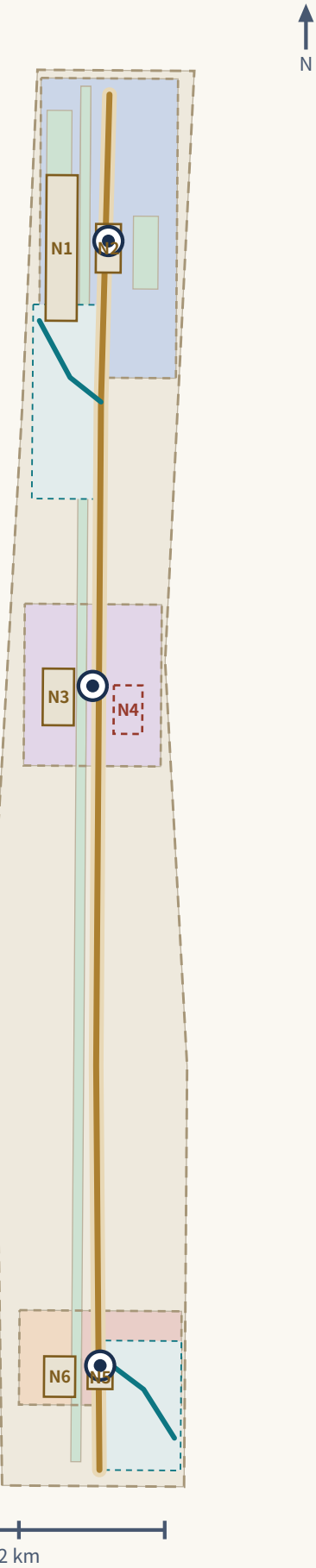
**Three positionings**  
Centennial Jing-Zhang cultural belt, urban AI everyday-life belt, AI convergence innovation belt.

**Five functions**  
A full-stack independent AI innovation system, a world-class AI innovation ecosystem, a new AI+ scenario-enablement paradigm, an intelligent and lively AI city, and a global voice in AI governance.

**Localized mechanism chain**  
Climate public space → research and talent anchor → a standing operating body → twin-city green compute cooperation → a digital twin with a rights boundary. The order is itself the design judgement: make the public space good without any device first, then layer the smart systems on.



Conceptual land-use partition



Conceptual land-use partition (closed, gapless, non-overlapping; areas recomputed in EPSG:4548)

| Feature | Name                                                                   | Code | Area     | Derivation                                                                                                                                             |
|---------|------------------------------------------------------------------------|------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| LU-001  | AI R&D innovation land (diagrammatic concept)                          | 0802 | 144.2 ha | PROV-KEY-001 minus the union of proposed green-space concept polygons that overlap it, clipped to SITE-001; computed and recomputed in EPSG:4548 with  |
| LU-002  | Community service facility land (diagrammatic concept)                 | 0702 | 88.2 ha  | PROV-KEY-002 minus the union of proposed green-space concept polygons that overlap it, clipped to SITE-001; computed and recomputed in EPSG:4548 with  |
| LU-003  | Commercial and enterprise service land (diagrammatic concept)          | 05   | 23.3 ha  | PROV-KEY-003 minus the union of proposed green-space concept polygons, split at longitude 116.347 so the market-node (N6) side is retained; clipped to |
| LU-004  | Cultural / railway heritage interpretation land (diagrammatic concept) | 0803 | 31.2 ha  | PROV-KEY-003 minus the union of proposed green-space concept polygons, split at longitude 116.347 so the heritage/crossing-node (N5) side is retained; |
| LU-005  | Park and open space land (diagrammatic concept)                        | 1401 | 129.5 ha | Union of all six submitted green-space concept polygons (GREEN-AXIS-001, GREEN-QINGHE-001, GREEN-XIAOYUE-001, GREEN-ST01-001, GREEN-ST02-001, GREEN-ST |
| LU-006  | Reversible reserve / flexible land (diagrammatic concept)              | 16   | 724.9 ha | SITE-001 minus the union of LU-001..LU-005; computed and recomputed in EPSG:4548 with a shared 1mm snapping grid so shared boundaries with adjacent la |

Regulatory depth in three tiers

- (1) Directly recomputable from the submitted geometry**

Land-use structure, green and public-space ratios, building footprint area. All are based on the provisional boundary and are for discussion only.
- (2) Controls requiring an official regulatory plan**

FAR, building height, density, setbacks, rights-of-way and facility standards. All are marked unknown or pending official conditions this round; no speculative value is passed off as an approved control.
- (3) Performance indicators needing operating data**

AI innovation index, talent density, active-travel accessibility, scenario usage and event participation enter annual review rather than becoming planning conclusions.

Four-tier building strategy and concept intervention envelopes

| Tier                           | Design action                                                      | Precondition                                                        |
|--------------------------------|--------------------------------------------------------------------|---------------------------------------------------------------------|
| Retain                         | Keep massing and frontage; maintenance only                        | Condition survey                                                    |
| Light renovation               | Activate ground floor, open frontage, tidy roof, add accessibility | Ownership and use consent                                           |
| Reversible addition            | Removable, relocatable lightweight structures                      | Fire, structural and planning permits                               |
| New build pending verification | Position intent only; no scale or height                           | All regulatory, ownership, utility and traffic conditions confirmed |

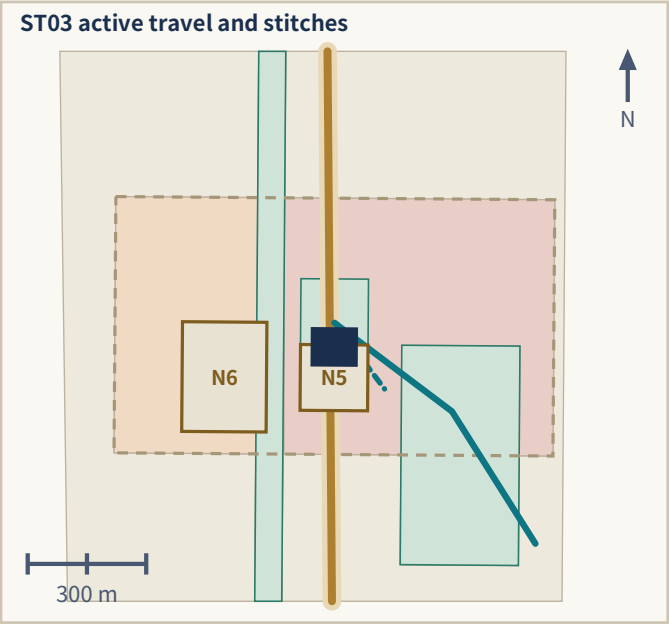
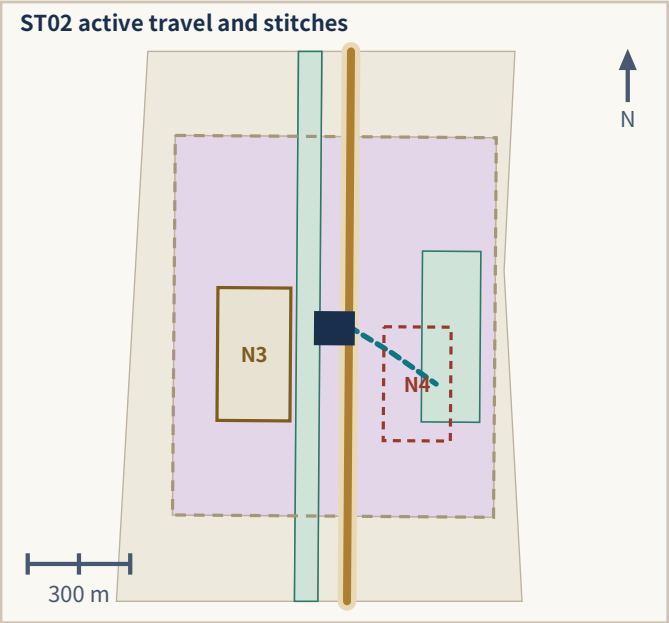
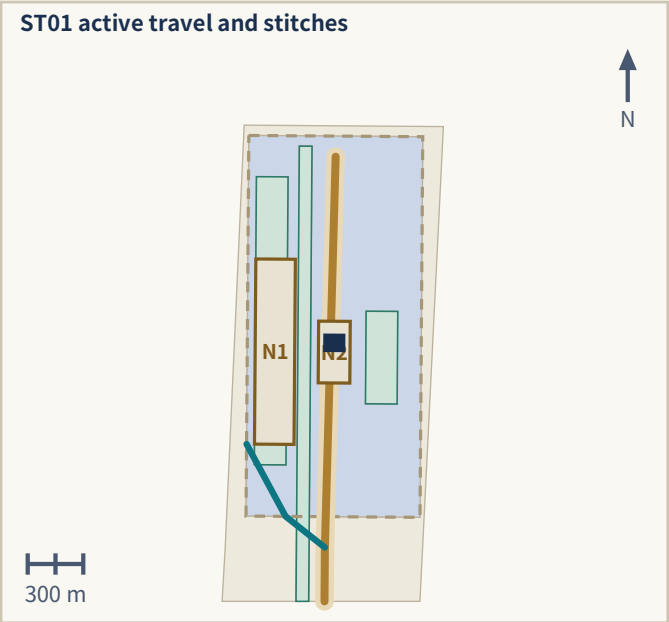
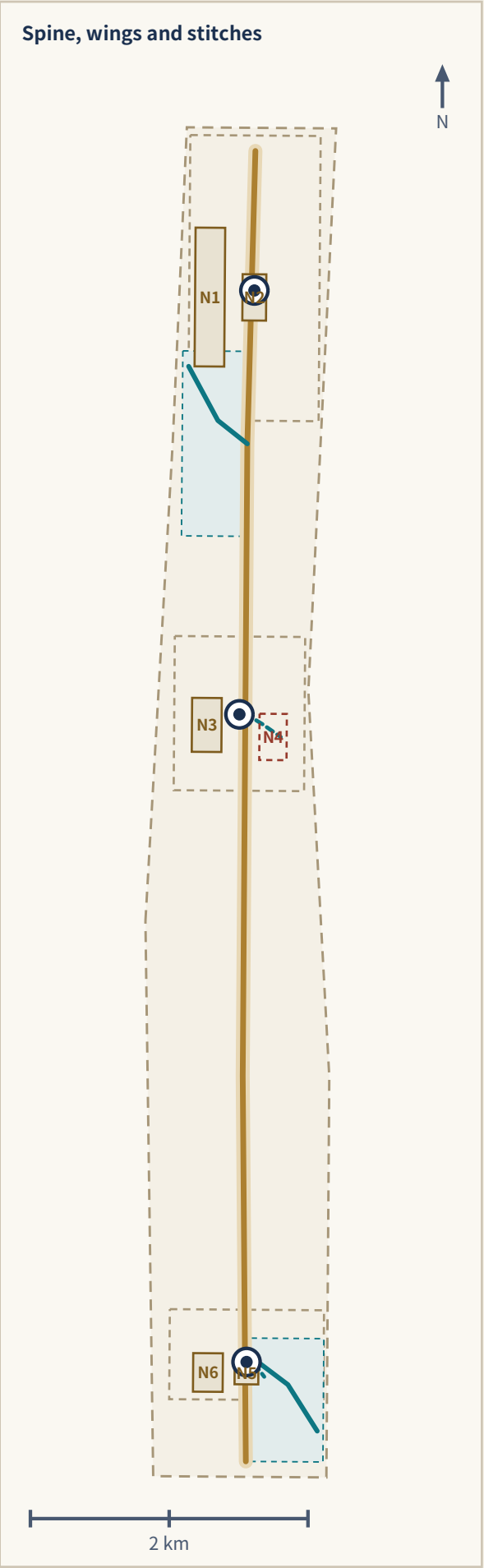
| Feature       | Name                                                     | Footprint                 |
|---------------|----------------------------------------------------------|---------------------------|
| BLDG-ST01-001 | Zhongzhiyuan Compute Exchange (conceptual envelope)      | 11,945.1 m <sup>2</sup>   |
| BLDG-ST02-001 | AI Origin Community Origin Commons (conceptual envelope) | 11,949.8 m <sup>2</sup>   |
| BLDG-ST03-001 | Dazhongsi Axis Launch Station (conceptual envelope)      | 11,957.0 m <sup>2</sup>   |
| Total         | building_footprint_area_sqm                              | 35,851.926 m <sup>2</sup> |

The three envelopes carry survey\_status=pending\_survey and are new concept intervention volumes, not existing footprints, GFA or FAR; floor\_area\_ratio remains unknown. No parcel-level retain/renovate/demolish conclusion is given.

# Mobility, active travel, stitching and utilities

Restore the east-west links cut by railway, ring roads and water, and make the spine continuously walkable

HAIDIAN AI AXIS: FROM COAL TO COMPUTE  
Railway Heritage. AI Innovation. Everyday Life.



## Five conceptual alignments (lengths recomputed in EPSG:4548; not engineering alignments)

| Feature               | Name                                                  | Class              | Nodes             | Length       |
|-----------------------|-------------------------------------------------------|--------------------|-------------------|--------------|
| ROAD-AXIS-001         | North-south public greenway spine (concept)           | greenway           | N1 N2 N3 N4 N5 N6 | 9,439 m      |
| ROAD-WING-TECH-001    | Zhongguancun technology-service wing link (concept)   | pedestrian         | N1 N2             | 714 m        |
| ROAD-WING-XIAOYUE-001 | Xiaoyue River scenario-enablement wing link (concept) | greenway           | N5 N6             | 768 m        |
| ROAD-STITCH-N4        | Campus-Station Stitch connector (concept)             | transit_connection | N4                | 306 m        |
| ROAD-STITCH-N5        | Four-Quadrant Crossing stitch (concept)               | pedestrian         | N5                | 210 m        |
| Total                 | conceptual_mobility_length_m                          |                    |                   | 11,437.434 m |

## N4: design\_only\_closed\_pending\_study

N4 Campus–Station Stitch: design\_only\_closed\_pending\_study. Drawn as a closed dashed outline, treated as not open and providing no through service; it asserts no bridge, tunnel, at-grade crossing, landing plaza or staffing decision.

## Utilities and new infrastructure: turning invisible compute into sited, accountable urban facilities

### Edge compute points

Placed in station-type spaces for low-latency public services; always co-located with rest, enquiry and charging, never as standalone machine rooms.

### Climate nodes

Combined with gardens and courtyards for shade, warmth, drinking water and temporary storm refuge; the physical carrier of the extreme-weather modes.

### City-side expression of failover

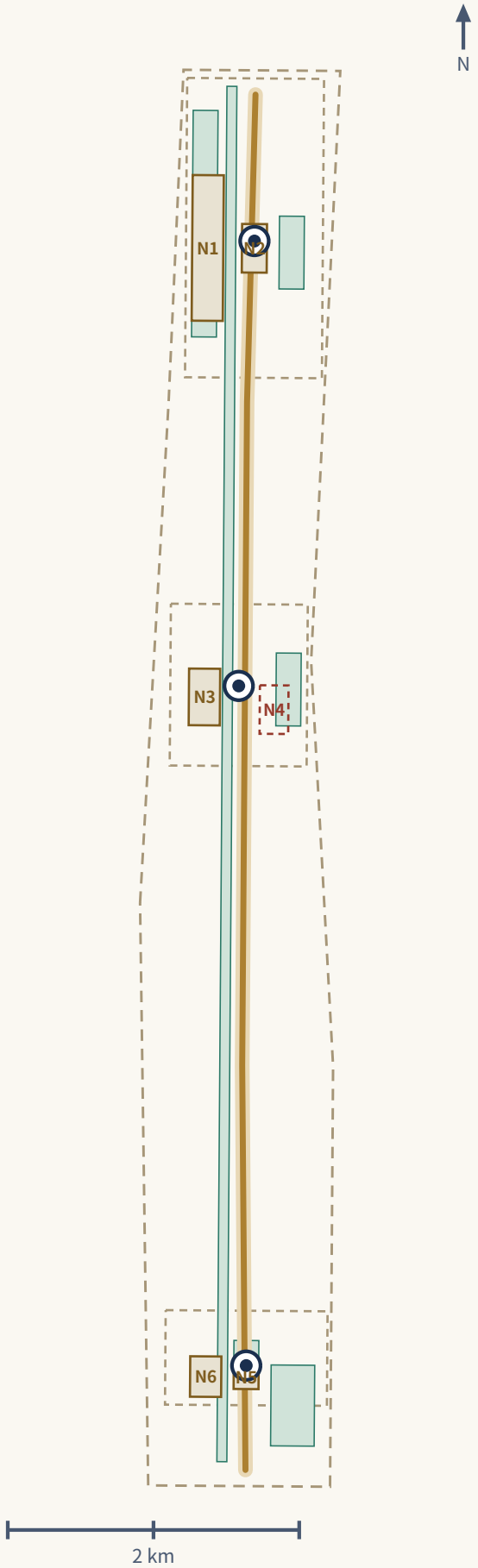
When an external link fails, stations display a degraded state and switch to staffed service; a black screen with nobody to ask is not acceptable.

### Conventional utilities

Existing water, drainage, power, telecom and sanitation capacity is unknown and is listed as a precondition for formal development; no load calculation is offered.

The CONSTRAINTS surface is a missing-data applicability and pre-implementation verification overlay. It is not service coverage and not a regulatory control line.

Blue-green network and heritage spine



Six blue-green concept surfaces (recomputed areas)

| Feature           | Name                                                | Area     |
|-------------------|-----------------------------------------------------|----------|
| GREEN-AXIS-001    | Heritage spine continuous shade corridor (concept)  | 64.5 ha  |
| GREEN-QINGHE-001  | Qinghe riverside detention green belt (concept)     | 26.5 ha  |
| GREEN-XIAOYUE-001 | Xiaoyue River seasonal refuge green space (concept) | 16.6 ha  |
| GREEN-ST01-001    | Zhongzhiyuan station commons green (concept)        | 8.5 ha   |
| GREEN-ST02-001    | AI Origin Community station commons green (concept) | 8.5 ha   |
| GREEN-ST03-001    | Dazhongsi station commons green (concept)           | 4.7 ha   |
| Union             | green_ratio = 0.113434                              | 129.5 ha |

Six public-space prototype nodes (recomputed areas)

| Node  | Name                          | Prototype                        | Area    |
|-------|-------------------------------|----------------------------------|---------|
| N1    | Qinghe Compute Commons        | Garden + Courtyard               | 21.3 ha |
| N2    | Fail-safe Test Yard           | Courtyard + Station              | 5.7 ha  |
| N3    | Origin Courtyard              | Courtyard + Gateway              | 8.3 ha  |
| N4    | Campus–Station Stitch         | Stitch + Station (pending study) | 6.5 ha  |
| N5    | Four-Quadrant Crossing        | Gateway + Station                | 2.8 ha  |
| N6    | New Axis Launch Market        | Market + Courtyard               | 5.9 ha  |
| Union | public_space_ratio = 0.044371 |                                  | 50.6 ha |

A climate-first public spine: everyday activity, storm detention, safe recovery

Everyday mode

Continuous shade, ventilation, seating and a rest/cooling point network; coverage spacing is a design target to be verified, currently unknown.

Storm-detention mode

Tiered storage and drainage, floodable green space and rain gardens; with no hydrology model, capacity can only be a design target.

Safe-recovery mode

Desilting, inspection and reopening; human takeover is the default fallback, not an exception.

Three AI pilgrimage landmarks and cultural narrative (concept)

Compute Exchange (Zhongzhiyuan)

Centred on a publicly readable dispatch and ledger interface that turns “where a task is computed” into visitable, questionable public content; the honour display shows public test records and evaluation reports, not company rankings.

Origin Commons (AI Origin Community)

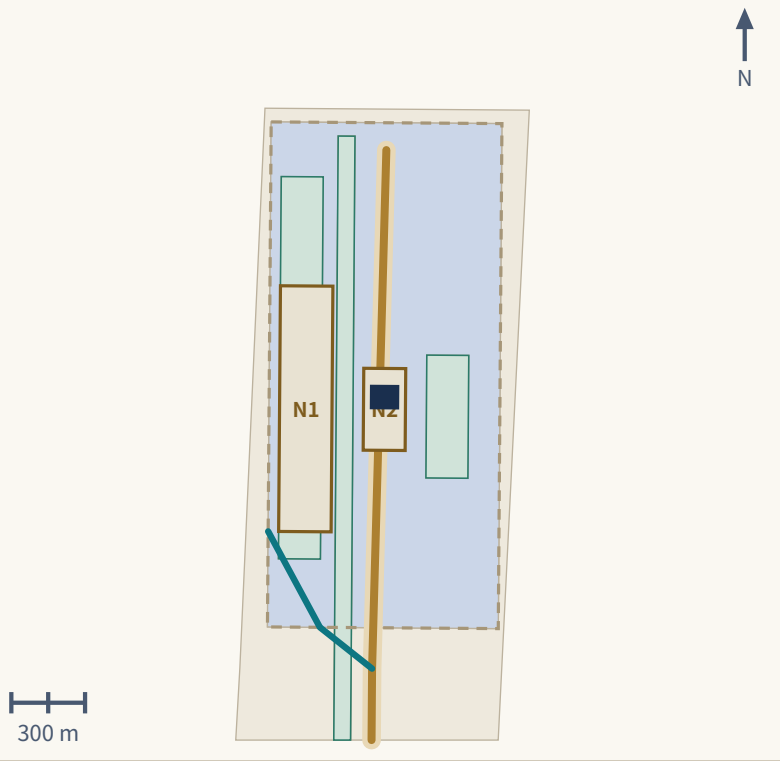
Centred on an open-source honour wall, community oral history and live model verification, recording contributors and contribution events; any display can be removed at a contributor's request.

Axis Launch Station (Dazhongsi)

Centred on the annual Axis Timetable and result launches while doubling as an everyday urban living room; international exchange and local everyday use share the same space.



ST01 concept spatial index



ST01 • Exchange and dispatch  
Zhongzhiyuan AI Independent Innovation Acceleration Area

Compute Exchange

PROV-KEY-001 • announced approx. 192.1 ha • recomputed 192.9 ha

- Positioning**  
The carrier of full-stack independent innovation and the voice in AI governance; the only node on the new axis that carries the decision and display of “where does the task get computed.”
- Spatial structure**  
The Qinghe-front interface becomes the public living room, forming a layered “river—garden—courtyard—R&D cluster” structure; dispatch and testing functions concentrate in a visitable central courtyard group, avoiding a closed command center.
- Building renewal direction**  
Primarily retention and light renovation, prioritizing ground-floor interfaces and rooftop equipment levels; new testing and display space preferentially uses reversible additions; new construction is a pending-verification item only after regulatory-plan and ownership conditions are obtained.
- Transport and walking-cycling**  
Strengthen separation of external traffic from internal walking and cycling; form continuous riding and walking paths along the Qinghe; keep human-staffed inquiry and non-digital entrances at connection points.
- Public space**  
The low-carbon innovation social environment centers on the Garden prototype, carrying shade, stormwater detention, and quiet rest; during public testing seasons some courtyards convert to open test grounds.
- AI scenarios**  
Scenario cards 02 Safety Governance Sandbox, 06 Jing-Zhang Heritage and AI Public Education, 08 Haidian–Ulanqab Green Compute Dispatch Test, 10 Climate Digital-Twin City Operations Test.
- Implementation risks**  
River blue lines and flood-protection conditions unknown; energy and safety requirements of dispatch facilities unknown; public opening and safety boundaries of test activities require dedicated study; complex ownership may prevent even reversible additions from landing.
- Project hooks**  
HX-02 Qinghe–Xiaoyue blue-green cross-link; HX-04 Compute Exchange and public ledger interface

N1 Qinghe Compute Commons

清河算力公园

Garden + Courtyard S06 S08 S10

Assumed existing conditions (pending survey verification)

The park edge along the Qinghe largely consists of boundaries and internal roads turned away from the river, with weak public access to the bank; at the same time “compute” is entirely invisible to the public, lacking a place to linger, watch, and ask questions.

Spatial actions

Reorganize the riverfront strip into a continuous linear park and embed a group of open courtyards within it, placing the publicly readable interface of dispatch and the ledger on the courtyards' street- and river-facing sides; link the courtyards with continuously shaded walkways — a combination of the Garden + Courtyard prototypes.

Day / night / extreme-weather modes

Day — a waterfront rest and viewing place freely open to the public; night — reduced illumination but continuous lighting and human-callable points retained, with the ledger interface switching to static display; extreme weather — courtyards become shade or rain-shelter nodes, and in storms the low-lying sections of the linear park operate as floodable green space with the corresponding walkways closed.

Main users

Park R&D and evaluation staff, nearby residents, riverside walkers and cyclists, public visitors.

Implementation preconditions

River blue lines and flood-protection standards, riverfront land ownership and maintenance responsibility, use consent for riverfront building-interface renovation, energy and safety requirements of the ledger interface.

Status

Concept node, indicatively located only within the provisional boundary; contains no coordinates, dimensions, engineering conclusions, or ownership judgments; open to deepening by professional teams.

Manual / offline fallback (declared geometry field, quoted verbatim)

Printed/static signage is on-site; a staffed help point is a design target during authorized service periods only, subject to confirmed operator, and is not a permanent-staffing claim. The dispatch ledger interface has an offline static-display fallback and never requires a digital identity to view.

N2 Fail-safe Test Yard

失效安全测试院

Courtyard + Station S02 S03

Assumed existing conditions (pending survey verification)

Safety evaluation and failure drills usually happen in closed interiors; the public has no way to judge what happens when a system fails, and nowhere to verify whether “human takeover” actually exists.

Spatial actions

Provide a hard-surfaced yard used daily as ordinary public ground, partially enclosed during test periods with movable barriers; run an outer gallery along the yard so the public can watch drills and human-takeover procedures without entering the test zone — a Courtyard + Station prototype.

Day / night / extreme-weather modes

Day — public ground or a bookable test yard; night — reverts to ordinary open ground with no drills scheduled; extreme weather — yields priority to shelter functions, tests are suspended, and the yard becomes temporary shelter and assembly space.

Main users

Evaluation and R&D teams, regulatory observers, public visitors, nearby residents (outside test periods).

Implementation preconditions

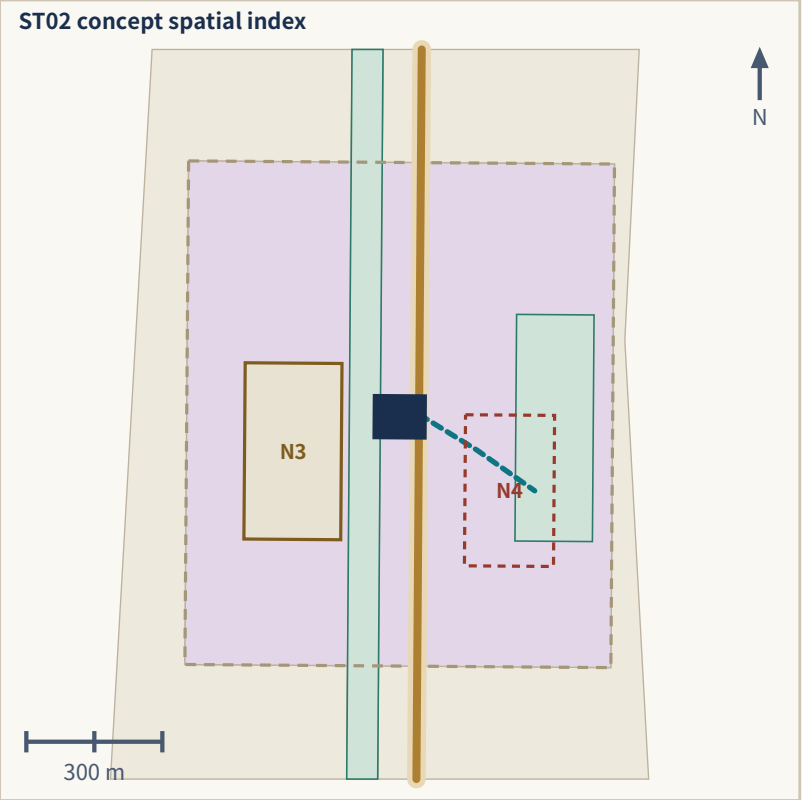
Public-safety boundaries and permits for test activities, site ownership and management responsibility, fire and structural conditions of barriers and facilities, confirmation of the rule that drills must not collect personal data.

Status

Concept node; form and operating mode are design suggestions, with no coordinates, dimensions, or engineering-feasibility conclusions.

Manual / offline fallback (declared geometry field, quoted verbatim)

Human takeover is the default action for any drill failure; printed instructions remain available on site. Staffed guidance is a design target during authorized drill/service periods only, subject to confirmed operator, with no digital step required.



ST02 • Verification and incubation  
Beijing AI Origin Community

Origin Commons

PROV-KEY-002 • announced approx. 104.3 ha • recomputed 104.3 ha

Positioning

The validation and incubation area of the world-class AI innovation ecosystem; models and public services undergo dual review by the community and universities here before entering the city.

Spatial structure

A small-scale network centered on the Courtyard prototype, stitching campus—park—neighborhood; open-source honor display is arranged linearly along the walking-and-cycling spine rather than concentrated in a single exhibition hall.

Building renewal direction

Primarily retention and light renovation, focusing on adding launch, talent-service, residential-support, and open-source collaboration space; prioritize reversible conversion of underused street-level ground floors.

Transport and walking-cycling

Walking-and-cycling stitching between campus and park comes first; night lighting and safe-route-home paths enter routine operations.

Public space

The Origin Commons must be genuinely enterable and lingerable with human-staffed service — no giant-screen hall; it hosts an information and service interface for newcomers' initial arrival period (service hours and staffing are design targets to be verified, subject to operating conditions).

AI scenarios

Scenario cards 01 Open-Source Launch Hall, 04 Newcomer 72-Hour Settling-In, 06 Jing-Zhang Heritage and AI Public Education, 09 Research-to-SME Product Conversion Test.

Implementation risks

Campus data and research outcomes need authorization; residential support involves property rights and existing community interests; community acceptance of test activities needs a long-term communication mechanism; without existing-building data, renovation classification can only be directional.

Project hooks

HX-03 Origin Commons and open-source honour display; HX-06 spine accessibility continuity and night-safety upgrade

N3 Origin Courtyard

原点庭院

Courtyard + Gateway S01 S06

Assumed existing conditions (pending survey verification)

Universities, enterprises, and residential communities are adjacent but lack a shared in-between place; open-source and model-validation activities mostly happen inside institutions, where residents can neither participate nor object.

Spatial actions

Organize a semi-enclosed courtyard as an in-between place open to all three parties, with street-level ground floors opened as enterable service and display interfaces; provide reservable small validation seats and a standing staffed service desk in the courtyard, with open-source honor display arranged linearly along the courtyard edge rather than concentrated in a hall — a Courtyard + Gateway prototype.

Day / night / extreme-weather modes

Day — shared use for courses, validation, launches, and everyday rest; night — lighting, seating, and human-callable points retained; validation activities end but the courtyard is not closed; extreme weather — the open ground floors become shaded / warming rest points, and validation yields to shelter use.

Main users

University students and faculty, developers, nearby residents (including older residents), renters newly arrived in Haidian, parents and children.

Implementation preconditions

Negotiation of campus boundaries and access management, use consent and business-mix adjustment for street-level ground floors, courtyard land ownership and maintenance responsibility, institutional arrangements for resident participation and objection channels.

Status

Concept node, a reference scheme open to deepening by professional teams, with no coordinates, dimensions, or findings on building condition.

Manual / offline fallback (declared geometry field, quoted verbatim)

Printed handbook available on site; a staffed service desk is a design target during authorized service periods only, subject to confirmed operator and annual authorization, not a permanent-staffing claim. Content review and any showcase can be withdrawn without notice, no login required.

N4 Campus–Station Stitch

校—站缝合节点

Stitch + Station (pending study) S04 S09

Current status (design\_only\_closed\_pending\_study)

This node is a mobility\_stitch prototype and a design intent only; until the dedicated transport, accessibility, and engineering studies are completed it is treated as not open and provides no passage service. Overpass, underpass, at-grade reconfiguration, a continuous sheltered segment, small landing plazas at both ends, detour organization, and staffing are all concept options pending study; this proposal presents none of them as a decided scheme.

Assumed existing conditions (pending survey verification)

Between campuses, parks, and rail stations there are detours created by railway, roads, or walls, forcing pedestrians and cyclists onto long circuitous paths, especially inconvenient at night and in rain or snow.

Spatial intent (all pending study)

A continuous, step-free campus–station walking-and-cycling stitch is this node's design target; the possible components — assembly and waiting space at both ends, shelter, seating and lighting, and the connector alignment — are put forward as concept options, a Stitch + Station prototype; the crossing form and every component can be fixed only after demonstration by professional teams.

Post-opening operating intent (design targets only)

Day — a path for school commutes, work commutes, and campus–park trips; night — lighting and visibility take priority; extreme weather — closure per contingency plan with alternate routes activated. Staffing and human guidance denote design targets only during authorized service windows after an operator is confirmed.

Main users (after the studies pass and the node opens)

University students and faculty, commuters, night-shift workers, students and parents, wheelchair users and others with limited mobility.

Implementation preconditions

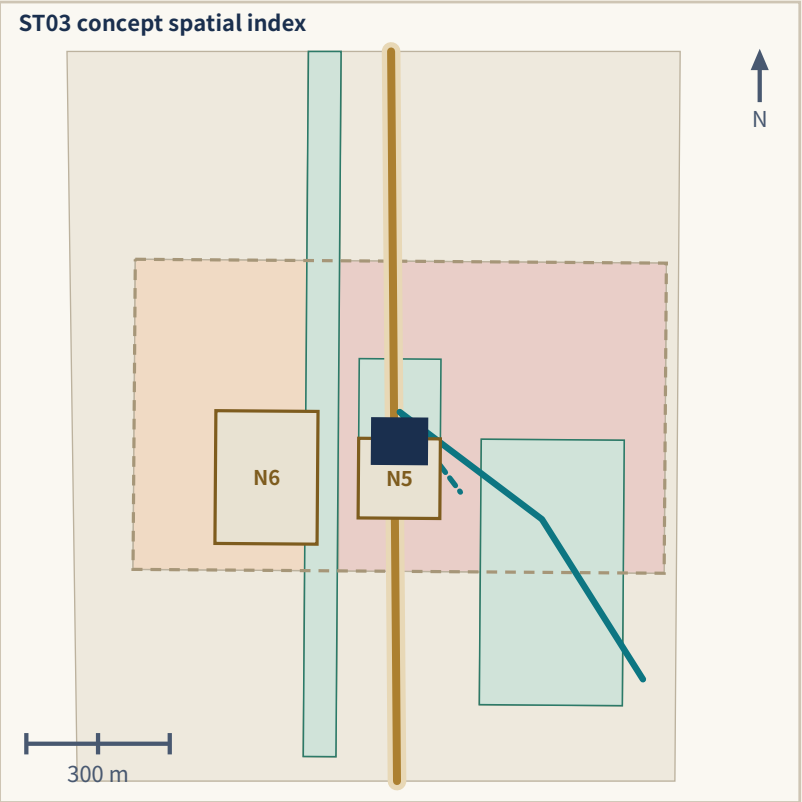
The dedicated transport, accessibility, and engineering studies; railway and road safety-control requirements; rail-station entrance conditions; ownership and municipal-utility conditions of the landing spaces at both ends.

Status

Concept node; form and location are indicative only and constitute no decision on bridge, underpass, at-grade crossing, sheltered segment, landing plaza, detour organization, or staffing, nor any alignment judgment or ownership arrangement.

Manual / offline fallback (declared geometry field, quoted verbatim)

No manual or staffed fallback is asserted. A continuous route and its fallback are design targets pending transport/accessibility/engineering study; until a feasible, authorized alternative is confirmed, N4 remains design-only and closed.



ST03 • Launch and translation

## Dazhongsi AI Industry Cluster

Axis Launch Station

PROV-KEY-003 • announced approx. 72.0 ha • recomputed 72.0 ha

**Positioning**  
The launch and conversion area for intelligence-native new business forms; the new axis's outward interface to the market, the public, and the world.

**Spatial structure**  
Led by the Market prototype, organizing four-quadrant pedestrian connectivity and ground-floor commercial services around the rail station; launch and cultural-content functions string along the pedestrian system.

**Building renewal direction**  
Primarily retention and ground-floor activation; compound use of planned green space supplements public space; new construction and additions are all pending-verification items subject to confirmation of regulatory-plan and ownership conditions.

**Transport and walking-cycling**  
Rail-station integration and four-quadrant pedestrian connectivity at the intersection are this area's most critical spatial actions; they require joint demonstration with rail, road, and municipal authorities, and this proposal offers no engineering conclusions.

**Public space**  
The Axis Launch Station is public space, not a corporate showroom; it is the main venue of the annual "New Axis Timetable" while remaining an everyday urban living room usable by ordinary citizens.

**AI scenarios**  
Scenario cards 05 Continuous Barrier-Free Passage, 07 Night-Shift Safe Journey Home, 03 Network/Power Outage Fail-Safe, 10 Climate Digital-Twin City Operations Test (launch end).

**Implementation risks**  
Rail and road engineering conditions unknown; four-quadrant connectivity involves intersection safety and existing utilities; numerous commercial owners blur public-space property rights and maintenance responsibility; compliance and copyright costs of international events are relatively high.

**Project hooks**  
HX-05 Dazhongsi four-quadrant pedestrian connection; HX-07 Axis Timetable and annual open test season

## N5 Four-Quadrant Crossing

四象限过街

Gateway + Station S05 S07

**Assumed existing conditions (pending survey verification)**  
Pedestrian connectivity between the four quadrants of the large intersection is unbalanced, with some directions requiring multiple waits or detours; the four corners are mostly leftover circulation space, lacking waiting and rest places where people can linger.

**Spatial actions**  
Reorganize the four corners into continuous, step-free walking surfaces and reserve sheltered waiting and rest space at each corner; drop-off, non-motorized parking, and rail-station entrances in all four quadrants are organized on the same walking surface — a Gateway + Station prototype. The specific crossing form must be demonstrated by traffic and engineering professionals; this proposal gives no conclusion.

**Day / night / extreme-weather modes**  
Day — the main walking surface for commuting, shopping, and rail transfer; night — lighting and visibility take priority, with human-callable points retained in waiting spaces; extreme weather — sheltered waiting spaces provide temporary rain, snow, and sun shelter; under ponding or icing, human guidance and alternate routes are activated.

**Main users**  
Rail commuters, nearby workers and visitors, older residents, wheelchair users and others with limited mobility, non-motorized vehicle users.

**Implementation preconditions**  
Intersection safety assessment and traffic-organization review, road red-line and municipal-utility conditions, connection with rail-station entrances, and delineation of ownership and management responsibility for the four corner sites.

**Status**  
Concept node; expresses pedestrian-connectivity intent only and constitutes no engineering conclusion on road alignment, crossing works, or rail connection.

**Manual / offline fallback (declared geometry field, quoted verbatim)**  
A non-digital, continuous ground-level route with no app or digital ticket required to cross is a design target, subject to professional transport/accessibility confirmation, an authorized service window, safety conditions, and annual authorization; no existing route is claimed. Staffed guidance is a design target during authorized service periods only, subject to confirmed operator and safety conditions, not a permanent-staffing claim.

## N6 New Axis Launch Market

新轴发布市集

Market + Courtyard S03 S10

**Assumed existing conditions (pending survey verification)**  
Launches, enterprise procurement, and cultural events mostly take place inside closed venues, leaving the space idle outside event days; meanwhile street-level ground floors lack an everyday usable public activity surface.

**Spatial actions**  
Use a permanent lightweight canopy to define a stretch of open street frontage, used daily as an ordinary market, rest, and display space, converting during launch seasons into launch and procurement grounds through demountable stalls and display boards; the canopy is permanent public infrastructure while event fittings are temporary — a Market + Courtyard prototype.

**Day / night / extreme-weather modes**  
Day — market, display, negotiation, and everyday rest; night — lighting and seating retained, converting to nighttime rest and small-event space, not closed for lack of events; extreme weather — the canopy provides shade and rain shelter; in strong wind or blizzard, temporary fittings are removed per contingency plan and events suspended.

**Main users**  
Nearby residents and workers, SMEs and startup teams, visitors and international guests, stall operators.

**Implementation preconditions**  
Ownership and use consent for the street frontage and land, public-event permits and safety plans, structural and fire conditions of the canopy, rights clearance and recycling arrangements for temporary fittings.

**Status**  
Concept node, a reference scheme combining operations and space, with no coordinates, dimensions, leasing arrangements, or construction commitments.

**Manual / offline fallback (declared geometry field, quoted verbatim)**  
Canopy and seating structure (design target) remains usable with power/network down; a printed contingency plan covers removal of temporary fixtures. Staffed presence during events is a design target subject to confirmed operator and authorized service periods, not a permanent-staffing claim.

Ten AI scenario cards sharing one rights boundary (human review, offline alternative, exit)

|                                                                                                                            |                                                                                                     |                                                                                                                                            |                                                                                                                                                        |                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><div>S01</div><div>Open-source Release Hall</div><div>Origin Community</div></div>                                    | <div><div>S02</div><div>Safety-governance Sandbox</div><div>Zhongzhiyuan</div></div>                | <div><div>S03</div><div>Network/power failure fail-safe</div><div>Stations and gateways, whole axis</div></div>                            | <div><div>S04</div><div>Newcomer 72-hour settling</div><div>Origin Community, gateways</div></div>                                                     | <div><div>S05</div><div>Continuous accessible movement</div><div>Entire spine</div></div>                                                         |
| <div><div>S06</div><div>Jing-Zhang heritage and AI public education</div><div>Heritage spine, Origin Community</div></div> | <div><div>S07</div><div>Safe night route for shift workers</div><div>Spine and stations</div></div> | <div><div>S08</div><div>Haidian–Ulanqab green compute dispatch test</div><div>Zhongzhiyuan</div><div>AI industry test scenario</div></div> | <div><div>S09</div><div>Research-to-SME product translation test</div><div>Origin Community, Dazhongsi</div><div>AI industry test scenario</div></div> | <div><div>S10</div><div>Climate digital-twin operations test</div><div>Whole axis, Qinghe–Xiaoyue</div><div>AI industry test scenario</div></div> |

Governance duties: types 1, 2 and 3 may not be held by one body

| Duty                                      | Scope and limits                                                                                                                 |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| 1 Public-interest trustee                 | Holds the free, continuous, enterable and non-digital-alternative floor; can veto any change that weakens basic public service   |
| 2 Technical operator                      | Day-to-day operation, maintenance and fault handling; does not self-assess or decide its own renewal                             |
| 3 Independent auditor                     | Verifies ledger definitions, energy/carbon/water metering and privacy enforcement; issues public audit opinions                  |
| 4 Residents' and users' council           | Represents residents, older people, mobility-restricted users, night workers and students; can demand review and move to suspend |
| 5 Haidian–Ulanqab agreement working group | Negotiates task tiering, metering definitions and fault liability; must never state that contracts or capacity are settled       |
| 6 Independent appeals and mediation body  | Receives complaints, mediates disputes and supervises suspension and exit                                                        |

Reversible authorization cycle and shared card rules

**90-day standards and no-personal-data drill (concept)**  
90-day standards and no-personal-data drill: rules, definitions, process and ledger table structure only; no personal data is collected at any point.

**One-year reversible public pilot (concept)**  
One-year reversible public pilot: trial operation at limited nodes; all facilities and rules are removable and revertible and can be stopped at any time.

**Annual renew, modify, or stop cycle (concept)**  
Annual audit and renew / modify / stop decision: the independent auditor issues a public opinion; stopping is a normal option.

Shared rules: every card must have a usable non-digital alternative under network, power or system failure, and every card must let a user opt out without losing basic public service. S08, S09 and S10 are the three AI industry test scenarios; until real operating data exist the ledger is published as a test specification only, with no fabricated measurements.



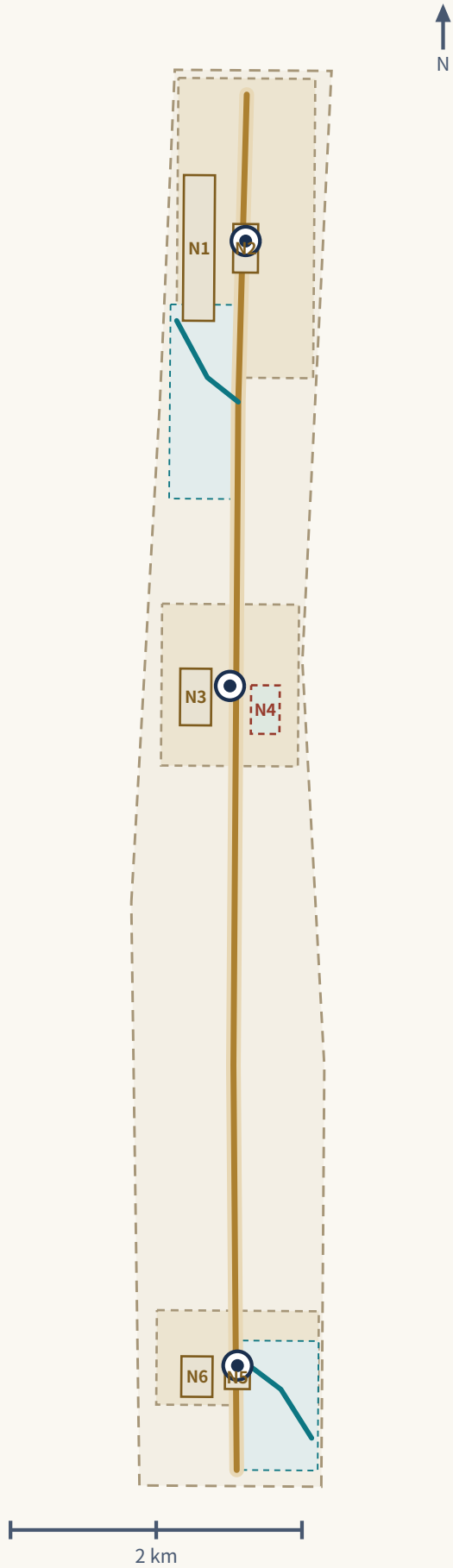
Renewal project list (concept only; no implementation or investment commitment)

| ID    | Project                                                  | Type                                | Key dependencies                                           | Key risks                                                       |
|-------|----------------------------------------------------------|-------------------------------------|------------------------------------------------------------|-----------------------------------------------------------------|
| HX-01 | Spine walking/cycling gap stitching and continuous shade | Public space / mobility             | Road right-of-way, space under bridges, traffic review     | Engineering conditions unknown; ring-road crossings are hardest |
| HX-02 | Qinghe–Xiaoyue blue-green cross-link and detention       | Blue-green space                    | River blue line, ecological and flood conditions           | No hydrology model; capacity can only be a design target        |
| HX-03 | Origin Commons and open-source honour display            | Urban renewal / culture             | Campus boundary, ownership, ground-floor uses              | Campus and community coordination takes a long time             |
| HX-04 | Compute Exchange and public ledger interface             | New infrastructure / public service | Energy, safety, operator and the Haidian–Ulanqab agreement | Without an agreement the ledger has no data to fill             |
| HX-05 | Dazhongsi four-quadrant pedestrian connection            | Rail integration / active travel    | Rail station, intersection, utilities                      | Involves several agencies and feasibility study                 |
| HX-06 | Spine accessibility continuity and night-safety upgrade  | Public space                        | Condition survey, lighting and maintenance body            | Without a survey neither target nor scope can be set            |
| HX-07 | Axis Timetable and annual open test season               | Operations / brand                  | Public-space permits, event safety, rights clearance       | No operator yet; continuity depends on long-term funding        |

Phasing logic: data governance → reversible pilot → three-station build → network extension → an

In the near term, do what does not depend on regulatory or engineering conditions: data and rights rules, reversible public-space pilots, ledger and test specifications. In the medium term, once conditions are confirmed, build the three stations. In the long term, extend the network laterally and institutionalize annual review. The open call's 100-day cycle is a submission deadline and is not the same thing as this implementation phasing.

Phasing layers (concept)



Three reversible authorization phases: each must be able to

| Feature   | Phase                                                 | Content                                                                                                                                                   | Declared flags                                 |
|-----------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| PHASE-001 | 10-day standards and no-personal-data drill (concept) | 90-day standards and no-personal-data drill: rules, definitions, process and ledger table structure only; no personal data is collected at any point.     | reversible=True<br>personal_data_allowed=False |
| PHASE-002 | One-year reversible public pilot (concept)            | One-year reversible public pilot: trial operation at limited nodes; all facilities and rules are removable and revertible and can be stopped at any time. | reversible=True<br>personal_data_allowed=False |
| PHASE-003 | Annual renew, modify, or stop cycle (concept)         | Annual audit and renew / modify / stop decision: the independent auditor issues a public opinion; stopping is a normal option.                            | reversible=True<br>personal_data_allowed=False |

Long-term operations and global events (agent.6)

**Annual events**  
The Axis Timetable is one calendar: the open test season (cards S08–S10), the open-source release season (S01), the heritage and public-education season (S06) and the annual ledger review; intensity follows season and climate.

**Scenario operation**  
The ten cards open through apply, assess, trial, review and exit; any card can be terminated without affecting basic public service.

**Attraction and translation**  
Auditable service performance (translation projects, exit rate, evaluation reports, participating teams) replaces capital-scale narrative; no investment, funding, policy or fiscal commitment is written in.

# Metrics, area recalculation and the evidence chain

Machine values, rounded display values and declaration coverage are stated separately

HAIDIAN AI AXIS: FROM COAL TO COMPUTE  
Railway Heritage. AI Innovation. Everyday Life.

## All 14 known metrics, source-synchronized (machine vs display value)

| Metric                                       | Machine value (metrics.json) | Display value            | Meaning and limit                                                                    |
|----------------------------------------------|------------------------------|--------------------------|--------------------------------------------------------------------------------------|
| site_area_sqm                                | 11412825.386 m²              | approx. 11.4 km²         | Recomputed area of the provisional rough boundary, not an official area              |
| building_footprint_area_sqm                  | 35851.926 m²                 | approx. 35,900 m²        | Sum of three concept intervention envelopes only; not existing footprint, GFA or FAR |
| green_ratio                                  | 0.113434                     | approx. 11.3%            | Union of six green concept surfaces ÷ provisional boundary area                      |
| public_space_ratio                           | 0.044371                     | approx. 4.4%             | Union of N1–N6 ÷ provisional boundary area                                           |
| conceptual_mobility_length_m                 | 11437.434 m                  | approx. 11.4 km          | Sum of five conceptual alignment lengths; not existing road length                   |
| key_area_count                               | 3                            | 3                        | Key areas                                                                            |
| station_count                                | 3                            | 3                        | Station envelopes                                                                    |
| wing_count                                   | 2                            | 2                        | Wings                                                                                |
| prototype_node_count                         | 6                            | 6                        | Prototype nodes                                                                      |
| scenario_count                               | 10                           | 10                       | Operating scenarios                                                                  |
| industry_test_count                          | 3                            | 3                        | Industry test scenarios                                                              |
| phase_count                                  | 3                            | 3                        | Phases                                                                               |
| declared_reversible_prototype_ratio          | 1.0                          | 100% (declaration basis) | Declaration / documentation coverage; not performance or effectiveness               |
| declared_manual_fallback_documentation_ratio | 1.0                          | 100% (declaration basis) | Declaration / documentation coverage; not performance or effectiveness               |

All 14 are low-confidence outputs from a provisional rough boundary and this package's concept geometry; they are not existing-condition data, regulatory controls, construction quantities, service performance or achieved KPI.

## Evidence chain: geometry → formula → metric → proposal citation

| Geometry evidence                              | Formula                             | Metric                       |
|------------------------------------------------|-------------------------------------|------------------------------|
| geometry/site_boundary.geojson#SITE-001        | polygon_area(SITE-001) @EPSG:4548   | site_area_sqm                |
| geometry/green_space.geojson#GREEN-* ×6        | union_area(GREEN-*) ÷ site_area_sqm | green_ratio                  |
| geometry/public_space.geojson#N1…N6            | union_area(N1..N6) ÷ site_area_sqm  | public_space_ratio           |
| geometry/roads.geojson#ROAD-* ×5               | Σ line_length(ROAD-*) @EPSG:4548    | conceptual_mobility_length_m |
| geometry/buildings.geojson#BLDG-ST01/02/03-001 | Σ polygon_area(BLDG-*)              | building_footprint_area_sqm  |

## Verifiable design targets: all currently unknown

| Design target                        | Control quantity and formula                                                | Value   |
|--------------------------------------|-----------------------------------------------------------------------------|---------|
| Rest / cooling point coverage        | Walking distance from any point on the spine to the nearest available point | unknown |
| Night availability of rest points    | Open hours / night-period hours                                             | unknown |
| Spine accessibility continuity       | Accessible continuous length / total spine length                           | unknown |
| Failure recovery time                | Time from degradation to recovery or completed human takeover               | unknown |
| Staff arrival and help-response time | Time from call to staff arrival or answer                                   | unknown |
| Fault repair duration                | Time from fault registration to completed repair                            | unknown |

## The 23 unknown metrics stay empty

accessibility\_service\_coverage\_ratio • current\_green\_ratio • current\_public\_space\_ratio • existing\_building\_area\_sqm • existing\_building\_density • existing\_building\_height\_m • fail\_safe\_pass\_rate • floor\_area\_ratio • haidian\_ulanqab\_dispatch\_capacity • haidian\_ulanqab\_energy\_carbon\_water\_performance • haidian\_ulanqab\_price • haidian\_ulanqab\_sla • hydrology\_capacity • investment\_return • jobs\_created • municipal\_utility\_capacity • official\_key\_area\_areas\_sqm • official\_site\_area\_sqm • participation\_outcomes • population\_coverage • project\_cost\_estimate • project\_schedule • retain\_renovate\_demolish\_area\_sqm

The two declared\_\* metrics describe declaration and documentation coverage in the submitted layer only. They are never performance, effectiveness or achieved KPI.



Factual and framing risk

- HAIDIAN AI AXIS is not a statutory planning axis, not an extension of or second Beijing Central Axis, not a World Heritage–related project, and not a government-approved name.
- The 1909 Jing-Zhang Railway connected Liucun in Fengtai, Beijing with Zhangjiakou and did not directly enter today's Inner Mongolia; that linkage belongs to the later Ping-Sui / Jing-Bao evolution chain.
- The Haidian–Ulanqab twin-city compute coordination is this proposal's auditable dispatch model (proposal inference), not an approved or operating official architecture; no capacity, price, SLA or investment is promised.
- Every staffed desk, staffed attendance or human-callable point is a design target realizable during authorized service windows after an operator is confirmed; it is not existing staffing.

This proposal claims no zero-carbon compute, no 4.2 ms direct link to Haidian and no Ulanqab-wide PUE; it promises no investment, capacity, SLA or funding. Middle-East case performance figures are project-side claims; only mechanisms are transferred.

Unresolved data gaps

- Official SITE\_BOUNDARY and the three KEY\_AREA polygons
- Regulatory plan sheets and controls (FAR, height, density, setbacks)
- Existing-building survey, ownership, structure and fire conditions
- Road rights-of-way, rail alignments and utility capacity
- River blue lines, flood-control standards and hydrology model
- Heritage protection lines and historic building inventory
- Accessibility survey and inventory of walking/cycling gaps
- Project-specific Haidian–Ulanqab agreement, SLA and operating ledger data

Data and geometry risk: the submitted boundary and the three key-area polygons are provisional, and rectangle edges are neither parcel lines nor rights-of-way. geometry/constraints.geojson is a missing-data applicability and pre-implementation verification overlay plus two wing catchments; it contains no statutory control line, and official control lines are still not part of the check. Without regulatory, existing-building, ownership, utility, fire and heritage conditions, every dependent conclusion is downgraded to a pending item and recorded in assumptions.json.

Privacy, human veto and copyright

Every AI scenario must keep human review, an offline alternative and exit conditions; a human can override an automated decision; personal identity, biometric, minor, health and trajectory data are never collected for outbound tasks; resident profiles are never used for commercial recommendation. If a scenario cannot meet these three, stop the scenario rather than lower the standard. The origin, licence and authorization status of every image, drawing, icon, font, dataset and code asset is registered in sources.json and report/copyright\_statement.md. The HTML pages load no remote script, map tile, font, iframe, form or external API and do not track reviewers.

Agreement-dependency risk

Every Haidian–Ulanqab dispatch, ledger and test scenario depends on a project-specific agreement. Until such an agreement, SLA or dataset is provided to and verified by this submission, the ledger is only an empty table structure with metering definitions, and the related cards exist only as test specifications.

How the boundary statements are placed

The boundary statements sit in exactly three places: the non-negotiable declarations opening chapter 1, the “do not copy / do not infer” column bound to each specific mechanism table, and this chapter's risk summary. Repetitions elsewhere are emphasis only. The more disclaimers repeat, the more readers skip them, diluting the three that matter.

Current package state

package\_state remains scaffold and the SCAFFOLD-DRAFT marker is retained. The complete Chinese and English bodies and 42 registered sources exist, and this package's concept geometry, metrics and assumptions have passed this round's internal review. The five bilingual figure pairs, bilingual HTML and A3/A0 PDF pairs are generated from the same source by this builder. Manifest listing and hash refresh, removal of the scaffold marker and package finalization remain for a later gated task; until then the package state does not change.

Self-check state (self\_check.json)

| Check                 | Result | Target                         |
|-----------------------|--------|--------------------------------|
| BOUNDARY_TRUST        | pass   | geometry/site_boundary.geojson |
| KEY_AREAS_TRUST       | pass   | geometry/key_areas.geojson     |
| LAND_USE_TOPOLOGY     | pass   | geometry/land_use.geojson      |
| VISUAL_STATIC         | pass   | visual/index.html              |
| PROFESSIONAL_EVIDENCE | pass   | proposal.md                    |

This proposal claims no official approval, approved regulatory plan, final land ownership, final construction scale or guaranteed implementation. Every spatial recommendation is a concept proposal for professional teams to develop further. Final judgement rests with people and professional teams.